higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROFESSIONAL ENGINEERING PRACTICE
NQF LEVEL 4

1 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1
1.1.1 D
1.1.2 I
1.1.3 G
1.1.4 E
1.1.5 M
1.1.6 C
1.1.7 H
1.1.8 A
1.1.9 L
1.1.10 K

(10 × 1) (10)

1.2
- The work output of the worker suddenly drops significantly, putting the other workers under pressure to complete tasks to meet deadlines.
- The worker suddenly fails to, or does not want to, communicate with other workers.
- The worker does not concentrate on the work at hand, and hence the quality of his or her work drops.
- The worker's character changes suddenly, becoming evasive or even aggressive towards others.
- The worker suddenly becomes more and more absent from work.

(5 × 1) (5)

1.3
1.3.1 As the factory will create a lot of smoke, it is necessary for the company to do research on the general wind direction,✓ to ensure that the final location of the factory will be such that the smoke will be blown away from the residential area.✓

1.3.2 The location of the factory must be such that it will not encroach on the habitat of the endangered bird species in the area.✓ If this cannot be completely accomplished, the location of the factory must ensure that no smoke will be blown over the eco-sensitive area.✓

1.3.3 The community as well as animals use the water from the river that flows past the town. The company must ensure that no effluent or any other industrial waste flows into the river,✓ as this will have an adverse effect on the health of the community, as well as animals that drink from the river.✓

(3 × 2) (6)

1.4
1.4.1 SAID
1.4.2 SAIMechE
1.4.3 ECSA
1.4.4 SAIEE

(4 × 1) (4)
[25]

QUESTION 2

2.1 2.1.1 B
2.1.2 A
2.1.3 C
2.1.4 C
2.1.5 A

(5 × 1) (5)

2.2 2.2.1 9 (1)

2.2.2 A rating factor is a rating out of a maximum of ten that the design team allocates to a concept against a specific criterion,✓ to indicate how well the concept satisfies the specific criterion.✓ (2)

2.2.3 8 (1)

2.2.4 36 (1)

2.2.5 Criterion total = 125 – (56 + 45) = 24
✓ ✓ ✓
Rating factor = 24/4 = 6 (4)
✓ ✓

2.2.6 Concept total = 32 + 72 + 49 = 164 (2)

2.2.7 Concept 2 (1)

2.2.8 Concept 2 had the highest concept total. (1)

2.3 2.3.1 Once the prototype has been built, the design team must test it against the specifications as determined by the customer.✓ This process will determine whether the design is suitable, or whether changes must be made.✓

2.3.2 Once the prototype has passed all the tests, the design team leader will report back to the customer with a technical report✓ as well as a verbal and visual presentation of the solution.✓

(2 × 2) (4)

2.4
- Physical model
- Mathematical model
- Simulation
- Scale model
- Mock-up
- Descriptive model
- Prescriptive model
- Industrial model

(Any 3 × 1) (3)

[25]

QUESTION 3

3.1 The manager must:
- know what she/he wants to say.
- speak loud enough so that everyone can hear him/her.
- avoid the use of words that are difficult to pronounce.
- not walk around a lot.
- avoid fidgeting.
- avoid hesitation and repetition.
- must not rush through the presentation.
- keep his/her voice lively and enthusiastic. (Any 6 × 1) (6)

3.2
- Line chart
- Column chart
- Bar chart
- Histogram
- Pie chart
- Block diagram
- Flow chart (Any 5 × 1) (5)

3.3 3.3.1 ABC Precision Engineering (1)

3.3.2 50✓ mm✓ (2)

3.3.3 120 mm (1)

3.3.4 Piercing punch (1)

3.3.5 First Angle Orthographic Projection (1)

3.3.6
- The Bolster is item number 5 on the drawing.
- There are 2 of them in total.
- It is made of mild steel. (3 × 1) (3)

3.4 3.4.1 False
3.4.2 True
3.4.3 False
3.4.4 True
3.4.5 True
(5 × 1) (5)
[25]

QUESTION 4

4.1
- Monitors
- Speakers
- CD/DVD writer
- Printers
- Plotters
- Projectors (Any 4 × 1) (4)

4.2 Central Processing Unit (1)

4.3 Computer programs such as Microsoft Excel✓ that run on computer hardware.✓ (2)

4.4
4.4.1 Java
4.4.2 machine
4.4.3 Interpreted
4.4.4 Compiled
4.4.5 Translated
(5 × 1) (5)

4.5 4.5.1
- Bold
- Italic
- Underline (3)

4.5.2 Column G indicates the highest totals across all four months✓ for each of the four spares sold.✓ (2)

4.5.3 = AVERAGE(B6:E6) (2)
✓ ✓

4.5.4 = D5 + D6 + D7 + D8 OR = (D5 + D6 + D7 + D8) (2)

4.6 4.6.1 Greater than or equal to

4.6.2 Not equal to
(2 × 1) (2)

4.7 4.7.1 ^

4.7.2 /
(2 × 1) (2)
[25]

TOTAL: 100

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